

MUHAMMAD AKBAR

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EDUCATION

NED University of Engineering and Technology

Bachelor of Science (B.S.) in Computer Science and Information Technology

Karachi, Pakistan

Oct 2022 – Jun 2026 (Expected)

Bahria College Karsaz

Pre-Engineering

Karachi, Pakistan

Aug 2020 – July 2022

EXPERIENCE

Integration Xperts

Data Science Intern

Karachi, Pakistan

June 2025 – August 2025

- Assisted in developing computer vision solutions, including factory compliance monitoring and a facial recognition-based access system, gaining experience in image analysis.
- Implemented automated testing using Pytest to evaluate age and gender classification models on CCTV image datasets, improving model accuracy.
- Created and managed Regions of Interest (ROIs) for retail use cases and integrated configurations into the project database, enhancing project efficiency.

Karachi Development Authority, IT Department

IT Intern

Karachi, Pakistan

March 2025 – April 2025

- Supported IT infrastructure maintenance and database management to ensure system reliability and data accuracy.
- Collaborated with the IT team on process documentation and system improvements within government IT operations.

PROJECTS

AI-Powered Financial Research Assistant | None

- Architected an agentic hybrid RAG system using FastAPI and LangGraph, combining unstructured semantic search (Pinecone) with structured XBRL financial fact retrieval (PostgreSQL) for accurate, reliable responses.
- Developed a semantic routing engine to classify user intent (quantitative vs. qualitative), dynamically grounding numerical queries in structured financial data to mitigate LLM hallucinations.
- Engineered a concurrent, just-in-time ETL pipeline using AsyncIO that detects missing SEC filings during live queries and triggers parallel scraping, parsing, and indexing (text and XBRL) for real-time responses.

AI-Powered Factory Compliance System | None

- Developed a real-time computer vision system using YOLOv8 to detect PPE violations, machine occupancy, worker flow anomalies, and cleaning events, demonstrating expertise in image analysis.
- Designed custom Regions of Interest (ROIs) with rule-based compliance tracking for precise, zone-specific monitoring.
- Integrated modular detectors (helmet, glove, occupancy, cleaning) within a scalable architecture, logging violations, occupancy durations, and shift-wise summaries for automated compliance reporting.

AI-Powered Face Recognition and Attendance System | None

- Developed a real-time facial recognition access system using MTCNN, FaceNet, and DeepSort with embedding-based matching via TensorFlow and DeepFace.
- Automated frame extraction, image augmentation, and database logging with OpenCV and OpenPyXL for efficient face-based access tracking.
- Designed a Streamlit dashboard for live monitoring and user registration, implementing modular OOP architecture with logging and exception handling.

TECHNICAL SKILLS

Programming Languages: Python, C++, C, SQL

AI/ML: Computer Vision, Natural Language Processing (NLP), Retrieval-Augmented Generation (RAG), Data Science

Technical Skills: Object-Oriented Programming, Data Structures & Algorithms, Backend API Development

Other: Problem Solving